

Financial Development and Income Classification: A Multivariate Analysis of 111 Economies

1 Introduction

The structure of global financial systems reflects deep-rooted differences in economic development, institutional quality, and growth trajectories. The World Bank classifies economies into income groups on the basis of per-capita GNI, yet financial development—measured across dimensions of depth, access, efficiency, and stability—is both a driver and a consequence of income. This report asks: *how well do financial development indicators characterise and differentiate economies at different income levels, and what latent structure underlies cross-country variation in financial systems?*

Answering this question requires handling substantial internal correlation among the 23 indicators—depth variables (private credit, bank assets, liquid liabilities) capture closely related constructs, and before- and after-tax profitability ratios are near-collinear—making dimensionality reduction a natural first step. I apply a four-stage pipeline: PCA addresses the dimensionality and collinearity problem in the indicator set; LDA on PCA scores answers the supervised question of income-group separability; hierarchical clustering provides an unsupervised cross-check on whether income groupings reflect genuine financial regimes; and non-metric MDS gives a distance-based map that does not impose linear structure.

2 Data

The analysis uses the Global Financial Development Database (GFDD; World Bank 2022), restricting to the 2019 cross-section. After selecting 23 indicators and removing economies with any missing value, the working dataset contains 111 countries spanning all four World Bank income groups—High (31), Upper-Middle (35), Lower-Middle (34), and Low (11)—and all seven geographic regions.

The 23 continuous indicators cover four theoretical dimensions: **Depth** (5 vars): private credit, bank assets, liquid liabilities, deposits, and domestic credit (all % of GDP). **Access** (2): bank branches and ATMs per 100,000 adults. **Efficiency** (9): net interest margin, overhead costs, ROA/ROE (before and after tax), cost-to-income, and government credit allocation. **Stability** (3): bank Z-score, credit-to-deposits ratio, and liquid assets ratio. Four further indicators capture concentration, deposits, remittances, and foreign claims. Variable scales differ substantially (Z-score ≈ 1 –55; liquid liabilities up to 600%+ of GDP), so all analyses use standardised data. The small Low-income group ($n = 11$) is handled in LDA by setting priors proportional to ob-

served group frequencies. Summary statistics appear in Table A1 of the Appendix.

3 Methods

PCA. Reduces the 23 standardised indicators to orthogonal components. The Kaiser criterion (eigenvalue > 1 ; Kaiser 1960) determines the number retained. A scree plot and biplot (PC1 vs. PC2) assess variance structure and correspondence to the four theoretical dimensions.

Discriminant Analysis. LDA is applied to PCA scores (not raw variables) to avoid instability from near-collinear pairs (ei05/ei09, ei06/ei10; di01/di02/di05) (Venables and Ripley, 2002). Assumptions are evaluated via within-group Mahalanobis Q-Q plots (Johnson and Wichern, 2007) and by comparing LOOCV accuracy against QDA, which relaxes equal-covariance.

Cluster Analysis. Ward’s minimum-variance linkage (Ward, 1963) on the standardised 23-variable matrix. Average silhouette width (Rousseeuw, 1987) over $k = 2, \dots, 6$ guides cluster-number selection.

NMDS. Non-metric MDS (Kruskal, 1964) (isoMDS, $k = 2$) on Manhattan distances, producing a 2-D map that preserves rank-order of dissimilarities rather than imposing linearity. This provides a genuinely distinct view from PCA, which is based on Euclidean covariance structure.

4 Results

4.1 Principal Component Analysis

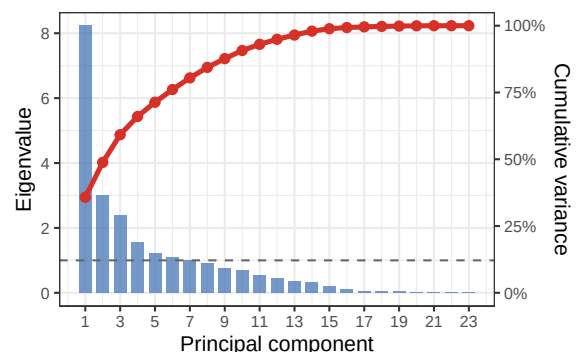


Figure 1: Scree plot with cumulative variance (red, right axis). Dashed line: eigenvalue = 1 (Kaiser threshold). 7 components retained, explaining 80.5% of variance.

The 7 retained components explain 80.5% of total variance. Table 1 shows loadings for the first four components. PC1 has large positive loadings on depth (di01, di02, di05, di08, di14) and access (ai02, ai25) variables, capturing the overall scale of financial intermediation.

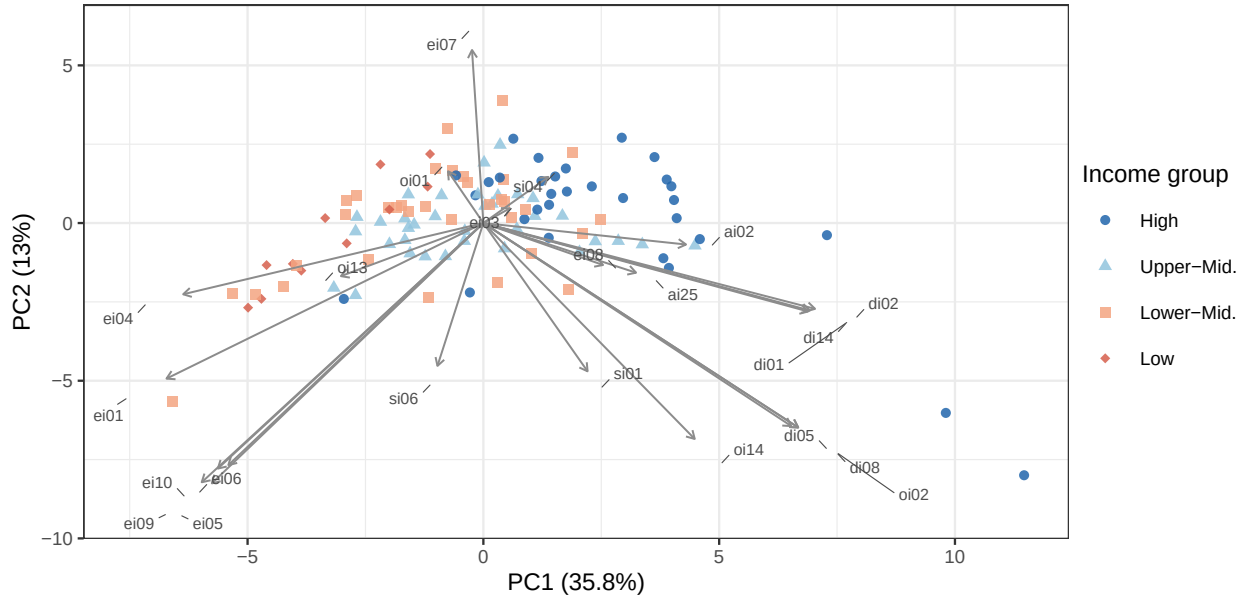


Figure 2: PCA biplot (PC1 vs. PC2). Points are coloured by income group; arrows show variable loadings. Depth and access variables (di*, ai*) point toward high PC1, where High-income economies concentrate. Low-income economies occupy the opposite end.

PC2 contrasts the cost-to-income ratio (ei07, positive) against bank profitability (ei05/ei06/ei09/ei10) and financial depth (di05/di08/oi02), capturing a cost-versus-return dimension within the banking sector.

Table 1: PCA loadings for the first four components.

	PC1	PC2	PC3	PC4
di01	0.291	-0.117	-0.226	-0.088
di02	0.296	-0.114	-0.233	-0.044
di05	0.275	-0.271	0.147	0.072
di08	0.281	-0.273	0.162	0.065
di14	0.290	-0.118	-0.213	-0.107
ai02	0.181	-0.029	0.137	-0.137
ai25	0.136	-0.066	-0.069	-0.489
ei01	-0.283	-0.207	-0.018	0.095
ei03	0.024	0.020	0.379	-0.316
ei04	-0.268	-0.095	0.147	0.039
ei05	-0.242	-0.347	-0.096	-0.106
ei06	-0.227	-0.323	-0.082	-0.187
ei07	-0.010	0.231	0.395	0.016
ei08	0.107	-0.056	-0.147	0.223
ei09	-0.251	-0.346	-0.091	-0.064
ei10	-0.237	-0.327	-0.084	-0.137
si01	0.093	-0.198	0.015	0.438
si04	0.058	0.062	-0.382	-0.131
si06	-0.041	-0.191	0.329	-0.145
oi01	-0.032	0.069	0.177	-0.281
oi02	0.281	-0.273	0.163	0.065
oi13	-0.127	-0.071	0.050	0.406
oi14	0.188	-0.288	0.297	0.067

4.2 Linear Discriminant Analysis

Figure 3 shows generally adequate normality for the High, Upper-Middle, and Lower-Middle groups. The Low-income group ($n = 11$) cannot reliably support a 7-dimensional covariance estimate, which is the structural cause of the strong departure from the χ^2 reference line—not random small-sample noise. This motivates the use of priors proportional to group frequencies in LDA. Comparing LOOCV accuracy: LDA achieves 54.1%, QDA achieves 53.2%. The similar accuracies indicate the equal-covariance assumption is not seriously violated; LDA is retained. However, this comparison should be treated with caution: QDA requires a separate within-group covariance estimate for each class; with only 11 Low-income observations, this estimate is

likely unreliable, so a lower QDA accuracy may reflect small-sample instability rather than genuine covariance heterogeneity.

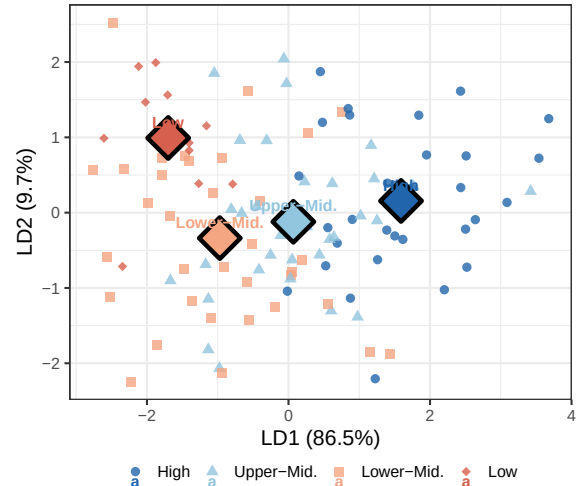


Figure 4: LD1-LD2 scatter plot. Filled diamonds mark group centroids. LD1 separates income groups in the expected order; LD2 captures residual within-group spread.

Table 2: LOOCV confusion matrix (rows = true, cols = predicted). Overall accuracy: 54.1%.

	Predicted			
	High	Upper-Mid.	Lower-Mid.	Low
High	20	11	0	0
Upper-Mid.	7	15	11	2
Lower-Mid.	3	7	20	4
Low	0	0	6	5

LD1 accounts for 86.5% of between-group variance, separating income groups in the expected order—High at the positive end, Low at the negative end. LOOCV accuracy is 54.1%, well above the naive baseline of 31.5% (always predicting the majority class). Note that PCA was fitted on all 111 observations before LOOCV; the cross-validation applies only to the LDA step, so accuracy is marginally optimistic, though the bias is small be-

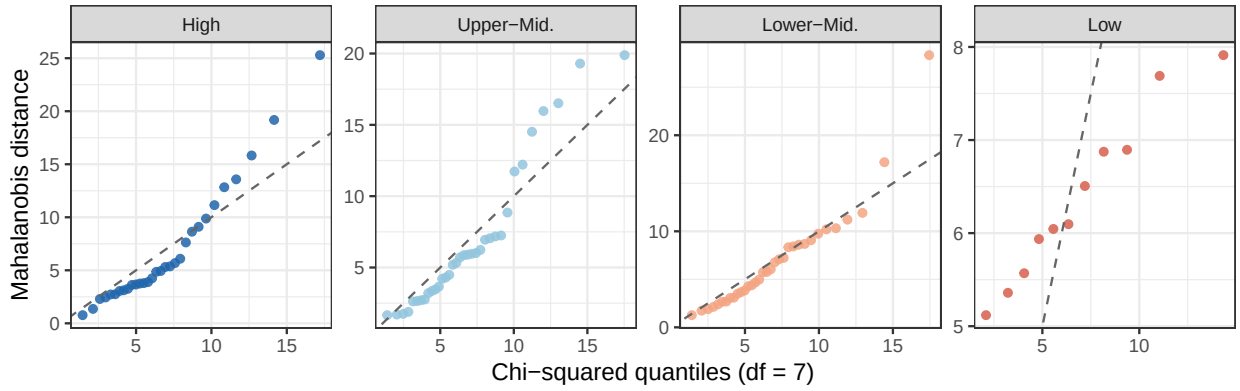


Figure 3: Within-group Mahalanobis distance Q-Q plots vs. χ^2_7 quantiles. Points near the dashed line indicate approximate multivariate normality. Mild heavy-tail departures visible in all groups; the Low-income group ($n = 11$) shows strong departure, reflecting structural over-parameterisation given only 11 observations and 7 PC dimensions.

cause PCA uses no class labels. Most misclassifications occur between adjacent income groups (Table 2), consistent with financial development representing a continuum rather than discrete regimes.

Given the concern raised by the Q-Q plot—that 7 PC dimensions cannot be reliably estimated from 11 Low-income observations—a sensitivity check was run using only the first four components (66% variance explained). LOOCV accuracy rises to 58.6%, slightly above the 7-component result, consistent with mild over-parameterisation under the Kaiser criterion. The substantive conclusions (income-group ordering on LD1, continuum pattern of misclassifications) are unchanged across both choices.

4.3 Cluster Analysis

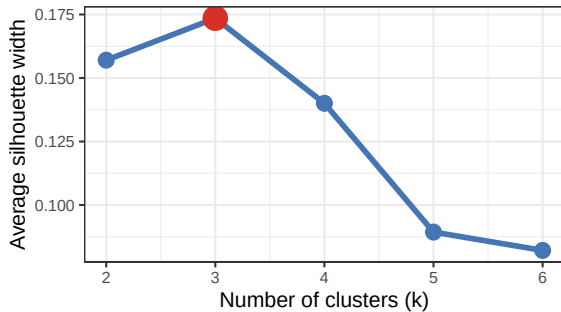


Figure 5: Average silhouette width for $k = 2, \dots, 6$ (Ward’s linkage). The peak at $k = 3$ (red point) is driven by HKG and LUX forming an isolated two-country cluster.

Hierarchical clustering uses the standardised 23-variable matrix directly. A parallel analysis on the retained PCA scores produced qualitatively identical results (same outlier cluster, same broad partitioning of income groups), so only the original-variable analysis is reported.

The silhouette curve (Figure 5) peaks at $k = 3$ (avg = 0.174), isolating Hong Kong (HKG) and Luxembourg (LUX) as a two-country cluster of extreme financial centres. Although the silhouette criterion favours $k = 3$, we report $k = 4$ because the three-cluster solution merely separates HKG and LUX into a trivially small outlier pair and provides no further differentiation among the remaining 109 economies; $k = 4$ yields a substantively more informative partition (Table 3, avg silhouette = 0.14). Cluster 2 ($n = 17$) groups advanced financial

Table 3: Ward clusters ($k = 4$) vs. World Bank income groups.

Cluster	High	Upper-Mid.	Lower-Mid.	Low
Cluster 1	17	14	21	4
Cluster 2	11	5	1	0
Cluster 3	1	16	12	7
Cluster 4	2	0	0	0

systems—predominantly High-income economies (Australia, Japan, Korea) alongside upper-middle income countries (China, Malaysia, Thailand). Clusters 1 and 3 partition remaining economies with substantial income-group mixing, echoing the LDA finding that adjacent groups are difficult to separate.

4.4 Non-metric MDS

The NMDS map (Figure 7, stress = 14.4%) preserves rank-order dissimilarities rather than linear distances, providing a non-linear complement to PCA. By Kruskal’s (1964) benchmark, stress below 20% indicates an acceptable fit; the obtained value of 14.4% falls in the “fair” range, meaning the 2-D configuration preserves the rank-order structure adequately but with some distortion. High-income economies still concentrate in one region and Low-income at the opposite end, but HKG and LUX appear even more isolated, confirming their outlier status. Singapore (SGP) is a moderate outlier within the High-income cluster.

4.5 Temporal Robustness: 2014 vs. 2019

The November 2021 GFDD release provides 2014 observations for the same 23 indicators, enabling a five-year temporal check. Of the 111 primary-analysis economies, 92 share complete 2014 records; the same PCA-LDA pipeline is applied to this matched panel.

The PCA covariance structure is highly stable: PC1 loadings correlate at $r = 0.984$ between years, and PC2 loadings at $r = 0.875$. Depth and access indicators dominate the first component in both years, confirming this is a structural feature of global financial development rather than a pattern specific to 2019.

LOOCV accuracy on the common 92 countries falls from 55.4% (2019, 6 components) to 46.7% (2014, 6 components, 77.9% variance explained; the Kaiser criterion retains one fewer component in the 92-country subset because the eigenvalue structure shifts slightly with the

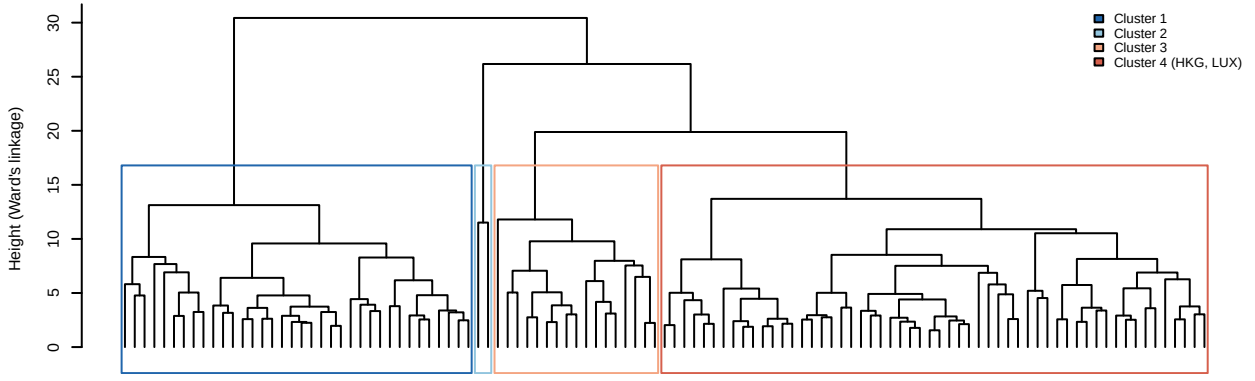


Figure 6: Ward's method dendrogram ($n = 111$). Rectangles delimit four clusters. Cluster 4 (HKG and LUX, far right) merges last and at large height, confirming their extreme outlier status.

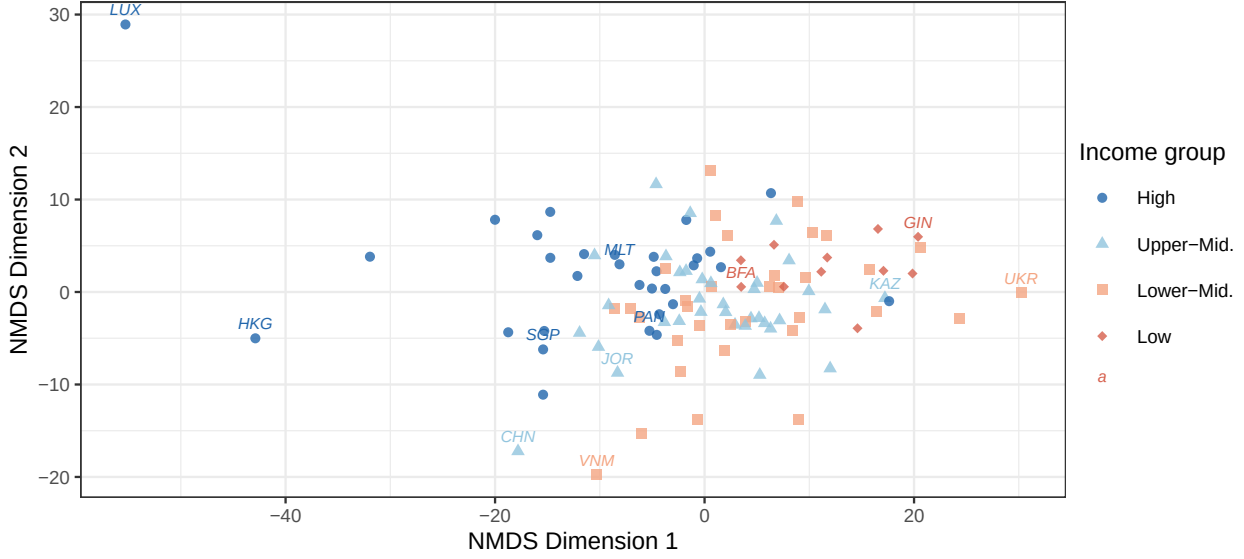


Figure 7: Non-metric MDS map of 111 countries (Manhattan distance; stress = 14.4%). ISO3 codes label known financial centres and the two most distant outliers per income group. HKG and LUX are isolated far from the main High-income cluster due to their disproportionately large financial sectors. LUX (extreme top position) compresses the vertical scale; the main cluster occupies $D_2 \in [-15, 15]$.

reduced sample). This modest decline suggests that income-group boundaries in financial-indicator space sharpened between 2014 and 2019, consistent with financial development divergence over this period. An alternative interpretation is that World Bank reclassifications between 2014 and 2019 brought income labels into closer alignment with countries' contemporaneous financial profiles, rather than a structural sharpening of financial regimes.

5 Conclusions

The 23 GFDD indicators carry substantial information about income-group membership: LOOCV accuracy of 54.1% compares favourably with the 31.5% naive baseline. The dominant axis of variation (PC1/LD1) captures financial depth and access, increasing monotonically from Low- to High-income economies. Most misclassifications occur between adjacent income groups (Table 2), consistent with financial development representing a continuum rather than discrete regimes.

Cluster analysis reveals that the dominant source of heterogeneity is the extreme outlier status of HKG and LUX. Beyond this outlier cluster, the remaining economies align only loosely with income groups (sil-

houette = 0.14 at $k = 4$), reinforcing the LDA finding that income groups represent a continuum rather than discrete financial regimes. The most striking structural finding is the isolation of Hong Kong and Luxembourg as an extreme outlier cluster—their financial sectors are so disproportionately large relative to their economies (foreign claims of 2014 exceeds 1000% of GDP for both) that they form a regime distinct from all other High-income economies. The NMDS map further identifies Singapore as a moderate outlier of the same type.

Limitations include restriction to bank-based indicators (omitting equity market depth and bond issuance) and the exclusion of economies with incomplete coverage across all 23 indicators. The temporal check (§4.5) partially addresses the cross-sectional limitation, though a full panel analysis covering multiple years would allow formal tests of financial convergence. Future work could apply robust discriminant methods or extend to market-based GFDD dimensions. PC-retention sensitivity and the QDA comparison are discussed in §4.2.

References

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- World Bank (2022). Global financial development database (GFDD).

Appendix: Summary Statistics and R Code

Table 4: Summary statistics for the 23 financial development indicators (unstandardised; $n = 111$ countries, 2019). All depth, efficiency, stability, and other variables are percentages unless noted.

Code	Dimension	Description	Mean	SD	Min	Max
di01	Depth	Private credit / GDP	53.87	39.67	7.66	236.76
di02	Depth	Bank assets / GDP	66.20	42.71	12.21	247.41
di05	Depth	Liquid liabilities / GDP	71.70	75.63	10.53	660.57
di08	Depth	Financial deposits / GDP	59.79	55.18	8.62	413.21
di14	Depth	Domestic credit / GDP	56.88	42.21	8.20	236.76
ai02	Access	Bank branches (per 100k)	16.97	13.45	0.42	64.48
ai25	Access	ATMs (per 100k)	55.65	53.84	1.62	328.07
ei01	Efficiency	Net interest margin	4.58	2.52	0.59	13.55
ei03	Efficiency	Noninterest income / total	35.44	9.97	13.84	63.54
ei04	Efficiency	Overhead costs / assets	3.06	1.61	0.57	7.14
ei05	Efficiency	ROA after tax	1.51	0.97	-0.98	6.09
ei06	Efficiency	ROE after tax	12.91	6.21	0.47	42.23
ei07	Efficiency	Cost-to-income	53.93	10.18	26.20	72.95
ei08	Efficiency	Govt credit / GDP	12.94	11.21	1.25	60.48
ei09	Efficiency	ROA before tax	1.96	1.23	-0.69	6.53
ei10	Efficiency	ROE before tax	16.66	7.75	0.96	45.27
si01	Stability	Bank Z-score	17.61	10.06	1.30	55.11
si04	Stability	Credit / deposits	100.36	71.01	21.40	677.24
si06	Stability	Liquid assets / deposits	33.87	14.52	8.79	70.95
oi01	Other	Bank concentration	63.68	18.05	16.14	100.00
oi02	Other	Bank deposits / GDP	59.76	55.15	8.62	413.21
oi13	Other	Remittances / GDP	5.03	6.35	0.00	27.97
oi14	Other	Foreign claims / GDP	44.01	112.30	0.04	1111.41

R Code

All analysis was conducted in R 4.5.1 (R Core Team, 2024). LDA, QDA, and NMDS used the MASS package (Venables and Ripley, 2002). Note: the data file carries a .csv extension but is structured as an Excel workbook (xlsx); `readxl::read_excel()` detects the format from the file header and reads it correctly. The code below reproduces every step from data loading to figure production.

```
library(knitr)
library(ggrepel)
opts_chunk$set(
  echo = FALSE,
  message = FALSE,
  warning = FALSE,
  fig.align = "center",
  dev = "pdf"
)
library(tidyverse)
library(readxl)
library(MASS)
library(cluster)
library(knitr)
library(kableExtra)

raw <- read_excel("Proposal/20220909-global-financial-development-database.csv",
  sheet = "Data - August 2022")

vars23 <- c(
  "di01", "di02", "di05", "di08", "di14",
  "ai02", "ai25",
  "ei01", "ei03", "ei04", "ei05", "ei06", "ei07", "ei08", "ei09", "ei10",
  "si01", "si04", "si06",
  "oi01", "oi02", "oi13", "oi14"
)

df2019 <- raw |>
  filter(year == 2019) |>
  dplyr::select(iso3, country, region, income, all_of(vars23)) |>
  drop_na() |>
  mutate(income = factor(income,
```

```

levels = c("High income","Upper middle income",
           "Lower middle income","Low income"))

X      <- scale(as.matrix(df2019[, vars23]))
income <- df2019$income

inc_labels <- c("High income"="High","Upper middle income"="Upper-Mid.",
               "Lower middle income"="Lower-Mid.,"Low income"="Low")
inc_colors <- c("High income"="#2166ac","Upper middle income"="#92c5de",
               "Lower middle income"="#f4a582","Low income"="#d6604d")
inc_shapes <- c("High income"=16L,"Upper middle income"=17L,
               "Lower middle income"=15L,"Low income"=18L)

dim_map <- c(di01="Depth",di02="Depth",di05="Depth",di08="Depth",di14="Depth",
            ai02="Access",ai25="Access",
            ei01="Efficiency",ei03="Efficiency",ei04="Efficiency",
            ei05="Efficiency",ei06="Efficiency",ei07="Efficiency",
            ei08="Efficiency",ei09="Efficiency",ei10="Efficiency",
            si01="Stability",si04="Stability",si06="Stability",
            oi01="Other",oi02="Other",oi13="Other",oi14="Other")

desc_lab <- c(
  di01="Private credit / GDP",    di02="Bank assets / GDP",
  di05="Liquid liabilities / GDP",di08="Financial deposits / GDP",
  di14="Domestic credit / GDP",  ai02="Bank branches (per 100k)",
  ai25="ATMs (per 100k)",        ei01="Net interest margin",
  ei03="Noninterest income / total",ei04="Overhead costs / assets",
  ei05="ROA after tax",          ei06="ROE after tax",
  ei07="Cost-to-income",         ei08="Govt credit / GDP",
  ei09="ROA before tax",         ei10="ROE before tax",
  si01="Bank Z-score",           si04="Credit / deposits",
  si06="Liquid assets / deposits",oi01="Bank concentration",
  oi02="Bank deposits / GDP",    oi13="Remittances / GDP",
  oi14="Foreign claims / GDP"
)

raw_mat <- as.matrix(df2019[, vars23])
desc_df <- data.frame(
  Code = vars23, Dimension = dim_map[vars23],
  Description = desc_lab[vars23],
  Mean = round(colMeans(raw_mat),2),
  SD   = round(apply(raw_mat,2,sd),2),
  Min  = round(apply(raw_mat,2,min),2),
  Max  = round(apply(raw_mat,2,max),2),
  row.names = NULL
)

pca <- prcomp(X, scale. = FALSE)
var_exp <- pca$sdev^2 / sum(pca$sdev^2)
cum_var <- cumsum(var_exp)
n_pc <- sum(pca$sdev^2 > 1)
scores_mat <- pca$x[, 1:n_pc, drop = FALSE]
lda_df <- data.frame(scores_mat, income = income)
prior_vec <- c(31, 35, 34, 11) / 111
lda_fit <- lda(income ~ ., data = lda_df, prior = prior_vec)
lda_cv <- lda(income ~ ., data = lda_df, prior = prior_vec, CV = TRUE)
lda_scores <- predict(lda_fit)$x
lda_acc <- mean(lda_cv$class == income)
lda_conf <- table(True = income, Predicted = lda_cv$class)
pct_ld <- round(lda_fit$svd^2 / sum(lda_fit$svd^2) * 100, 1)
qda_cv <- qda(income ~ ., data = lda_df, prior = prior_vec, CV = TRUE)
qda_acc <- mean(qda_cv$class == income)
lda_vs_qda_text <- if (qda_acc > lda_acc + 0.02) {
  "The higher QDA accuracy suggests meaningful covariance heterogeneity; QDA is preferred."
}

```

```

} else {
  "The similar accuracies indicate the equal-covariance assumption is not seriously violated; LDA is ret.
}
# sensitivity check: LDA on first 4 PCs
lda_cv4 <- lda(income ~ ., data = data.frame(pca$x[,1:4], income=income),
              prior = prior_vec, CV = TRUE)
lda_acc4 <- mean(lda_cv4$class == income)
pc_cols <- paste0("PC", 1:n_pc)
mah_list <- lapply(levels(income), function(grp) {
  sub <- as.matrix(lda_df[lda_df$income == grp, pc_cols])
  mah <- mahalanobis(sub, colMeans(sub), cov(sub))
  n <- length(mah)
  data.frame(income=grp, observed=sort(mah),
             expected=qchisq(ppoints(n), df=n_pc))
})
mah_df <- do.call(rbind, mah_list)
mah_df$income <- factor(mah_df$income, levels = levels(income))
# Euclidean distance for Ward clustering; Manhattan distance used separately for NMDS
d_euc <- dist(X, method = "euclidean")
hc <- hclust(d_euc, method = "ward.D2")
k4 <- cutree(hc, k = 4)
sil_vals <- sapply(2:6, function(k) mean(silhouette(cutree(hc,k), d_euc)[,3]))
sil_avg <- sil_vals[3]
cluster_tab <- table(Cluster = k4, 'Income group' = income)
# sanity check: clustering on PCA scores gives same result as on standardised variables
d_pcs <- dist(scores_mat, method = "euclidean")
hc_pcs <- hclust(d_pcs, method = "ward.D2")
k4_pcs <- cutree(hc_pcs, k = 4)
table(k4_pcs, k4) # should show high agreement
# non-metric MDS on Manhattan distances
d_manh <- dist(X, method = "manhattan")
set.seed(767)
nmads <- isoMDS(d_manh, k = 2, trace = FALSE)
nmads_stress <- nmads$stress
mds_df <- data.frame(
  D1 = nmads$points[,1],
  D2 = nmads$points[,2],
  iso3 = df2019$iso3,
  income = income
)
# load November 2021 release; use 2014 cross-section for temporal comparison
raw21 <- read_excel("Proposal/20220909-global-financial-development-database.csv",
                   sheet = "Data - November 2021")
df2014 <- raw21 |>
  filter(year == 2014) |>
  dplyr::select(iso3, country, region, income, all_of(vars23)) |>
  drop_na() |>
  mutate(income = factor(income,
                        levels = c("High income", "Upper middle income",
                                   "Lower middle income", "Low income")))
common_iso <- intersect(df2019$iso3, df2014$iso3)
df19c <- df2019 |> filter(iso3 %in% common_iso) |> arrange(iso3)
df14c <- df2014 |> filter(iso3 %in% common_iso) |> arrange(iso3)
X19c <- scale(as.matrix(df19c[, vars23]))
X14c <- scale(as.matrix(df14c[, vars23]))
income19c <- df19c$income
income14c <- df14c$income
pca19c <- prcomp(X19c, scale. = FALSE)
pca14c <- prcomp(X14c, scale. = FALSE)
pc1_cor <- round(cor(pca19c$rotation[,1], pca14c$rotation[,1]), 3)
pc2_cor <- round(cor(pca19c$rotation[,2], pca14c$rotation[,2]), 3)

```

```

n_pc14 <- sum(pca14c$sdev^2 > 1)
cum14 <- cumsum(pca14c$sdev^2 / sum(pca14c$sdev^2))
prior14 <- as.numeric(table(income14c)) / length(income14c)
lda_cv14 <- lda(income ~ .,
               data = data.frame(pca14c$x[, 1:n_pc14], income = income14c),
               prior = prior14, CV = TRUE)
lda_acc14 <- round(mean(lda_cv14$class == income14c) * 100, 1)

n_pc19c <- sum(pca19c$sdev^2 > 1)
prior19c <- as.numeric(table(income19c)) / length(income19c)
lda_cv19c <- lda(income ~ .,
               data = data.frame(pca19c$x[, 1:n_pc19c], income = income19c),
               prior = prior19c, CV = TRUE)
lda_acc19c <- round(mean(lda_cv19c$class == income19c) * 100, 1)
scree_df <- data.frame(PC=seq_len(23), Eigenvalue=pca$sdev^2, CumVar=cum_var)
max_eig <- max(scree_df$Eigenvalue)
ggplot(scree_df, aes(x=PC)) +
  geom_col(aes(y=Eigenvalue), fill="#4575b4", alpha=0.75, width=0.7) +
  geom_line(aes(y=CumVar*max_eig), colour="#d73027", linewidth=0.9) +
  geom_point(aes(y=CumVar*max_eig), colour="#d73027", size=1.5) +
  geom_hline(yintercept=1, linetype="dashed", colour="grey40") +
  scale_y_continuous(name="Eigenvalue",
                    sec.axis=sec_axis(~./max_eig, name="Cumulative variance",
                                       labels=scales::percent_format(accuracy=1))) +
  scale_x_continuous(breaks=seq(1,23,by=2)) +
  labs(x="Principal component") +
  theme_bw(base_size=8)
load_mat <- round(pca$rotation[,1:4], 3)
colnames(load_mat) <- paste0("PC",1:4)
kable(load_mat, format="latex", booktabs=TRUE,
      caption="PCA loadings for the first four components.",
      label="loadings", linesep="") |>
  kable_styling(latex_options=c("hold_position","scale_down"), font_size=6.5)
scores_df2 <- as.data.frame(pca$x[,1:2]); scores_df2$income <- income
loads_df2 <- as.data.frame(pca$rotation[,1:2]); loads_df2$variable <- rownames(loads_df2)
sf <- (max(abs(as.matrix(scores_df2[,c("PC1","PC2")]))) /
      max(abs(as.matrix(loads_df2[,c("PC1","PC2")])))) * 0.72
ggplot() +
  geom_point(data=scores_df2, aes(PC1,PC2,colour=income,shape=income),
            size=1.5, alpha=0.85) +
  geom_segment(data=loads_df2,
              aes(x=0,y=0,xend=PC1*sf,yend=PC2*sf),
              arrow=arrow(length=unit(0.12,"cm")), colour="grey55", linewidth=0.35) +
  geom_text_repel(data=loads_df2,
                 aes(x=PC1*sf*1.12, y=PC2*sf*1.12, label=variable),
                 size=2.0, colour="grey30",
                 max.overlaps=Inf, segment.size=0.2, min.segment.length=0.1) +
  scale_colour_manual(values=inc_colors, labels=inc_labels, name="Income group") +
  scale_shape_manual(values=inc_shapes, labels=inc_labels, name="Income group") +
  labs(x=paste0("PC1 (",round(var_exp[1]*100,1),"%)" ),
       y=paste0("PC2 (",round(var_exp[2]*100,1),"%)" )) +
  theme_bw(base_size=9) +
  theme(legend.position="right", legend.margin=margin(l=-4))
ggplot(mah_df, aes(expected, observed)) +
  geom_point(aes(colour=income), size=1.2, alpha=0.85) +
  geom_abline(slope=1, intercept=0, linetype="dashed", colour="grey40") +
  facet_wrap(~income, scales="free",
            labeller=labeller(income=inc_labels), nrow=1) +
  scale_colour_manual(values=inc_colors, guide="none") +
  labs(x=paste0("Chi-squared quantiles (df = ",n_pc,")"),

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    y="Mahalanobis distance") +
  theme_bw(base_size=9)
ld_df <- data.frame(LD1=-lda_scores[,1], LD2=lda_scores[,2], income=income)
centroids <- ld_df |> group_by(income) |>
  summarise(LD1=mean(LD1), LD2=mean(LD2), .groups="drop")
ggplot(ld_df, aes(LD1,LD2,colour=income,shape=income)) +
  geom_point(size=1.4, alpha=0.80) +
  geom_point(data=centroids, aes(fill=income),
    colour="black", size=5, shape=23, stroke=1.2) +
  scale_fill_manual(values=inc_colors, guide="none") +
  geom_text(data=centroids,
    aes(label=inc_labels[as.character(income)]),
    vjust=-0.9, size=2.2, fontface="bold") +
  scale_colour_manual(values=inc_colors, labels=inc_labels, name=NULL) +
  scale_shape_manual(values=inc_shapes, labels=inc_labels, name=NULL) +
  labs(x=paste0("LD1 (",pct_ld[1],"%)" ),
    y=paste0("LD2 (",pct_ld[2],"%)" )) +
  theme_bw(base_size=8) +
  theme(legend.position="bottom", legend.key.size=unit(0.3,"cm"),
    legend.margin=margin(t=-4))
conf_df <- as.data.frame.matrix(lda_conf)
rownames(conf_df) <- inc_labels[rownames(conf_df)]
colnames(conf_df) <- inc_labels[colnames(conf_df)]
kable(conf_df, format="latex", booktabs=TRUE, linesep="",
  caption=paste0("LOOCV confusion matrix (rows = true, cols = predicted). Overall accuracy: ",
    round(lda_acc*100,1),"%"),
  label="confusion") |>
  add_header_above(c(" " =1, "Predicted"=4)) |>
  kable_styling(latex_options=c("hold_position","scale_down"), font_size=7)
sil_df <- data.frame(k=2:6, AvgSil=sil_vals)
ggplot(sil_df, aes(k, AvgSil)) +
  geom_line(colour="#4575b4", linewidth=1) +
  geom_point(colour="#4575b4", size=2.5) +
  geom_point(data=sil_df[which.max(sil_df$AvgSil),],
    colour="#d73027", size=4) +
  scale_x_continuous(breaks=2:6) +
  labs(x="Number of clusters (k)", y="Average silhouette width") +
  theme_bw(base_size=8)
ct <- as.data.frame.matrix(cluster_tab)
colnames(ct) <- inc_labels[colnames(ct)]
ct <- cbind(Cluster = paste0("Cluster ", rownames(ct)), ct)
rownames(ct) <- NULL
kable(ct, format="latex", booktabs=TRUE, linesep="",
  caption="Ward clusters ($k=4$) vs.\\ World Bank income groups.",
  label="cluster", row.names=FALSE) |>
  kable_styling(latex_options=c("scale_down"), font_size=8)
par(mar=c(1.5,3.8,0.3,0.5), cex=0.5)
plot(hc, labels=FALSE, main="", xlab="", sub="",
  ylab="Height (Ward's linkage)", hang=-1)
rect.hclust(hc, k=4, border=c("#2166ac", "#92c5de", "#f4a582", "#d6604d"))
legend("topright",
  legend=c("Cluster 1","Cluster 2","Cluster 3","Cluster 4 (HKG, LUX)"),
  fill=c("#2166ac", "#92c5de", "#f4a582", "#d6604d"),
  bty="n", cex=0.75)
centroid_nmds <- mds_df |> group_by(income) |>
  summarise(cD1=mean(D1), cD2=mean(D2), .groups="drop")
mds_df2 <- mds_df |>
  left_join(centroid_nmds, by="income") |>
  mutate(dist_c=sqrt((D1-cD1)^2+(D2-cD2)^2))
fc_iso <- c("LUX","HKG","SGP","CHE","PAN","MLT","CYP","JOR")
top_out <- mds_df2 |> group_by(income) |> slice_max(dist_c, n=2) |> ungroup()

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label_df <- bind_rows(top_out, mds_df2 |> filter(iso3 %in% fc_iso)) |>
  distinct(iso3, .keep_all=TRUE)
ggplot(mds_df2, aes(D1,D2,colour=income,shape=income)) +
  geom_point(size=1.5, alpha=0.80) +
  geom_text(data=label_df, aes(label=iso3,
    size=2.0, vjust=-0.8, fontface="italic") +
  scale_colour_manual(values=inc_colors, labels=inc_labels, name="Income group") +
  scale_shape_manual(values=inc_shapes, labels=inc_labels, name="Income group") +
  labs(x="NMDS Dimension 1", y="NMDS Dimension 2") +
  theme_bw(base_size=9) +
  theme(legend.position="right", legend.margin=margin(l=-4))
kable(desc_df, format="latex", booktabs=TRUE,
  caption="Summary statistics for the 23 financial development indicators (unstandardised; $n=111$ c",
  label="summary",
  col.names=c("Code", "Dimension", "Description", "Mean", "SD", "Min", "Max"),
  linesep="") |>
  kable_styling(latex_options=c("hold_position", "scale_down"), font_size=8)

```